

VMDNet: Temporal Leakage-Free Variational Mode Decomposition for Electricity Demand Forecasting

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Abstract—Accurate electricity demand forecasting is challenging due to the strong multi-periodicity of real-world demand series, which makes effective modeling of recurrent temporal patterns crucial. Decomposition techniques make such structure explicit and thereby improve predictive performance. Variational Mode Decomposition (VMD) is a powerful signal-processing method for periodicity-aware decomposition and has seen growing adoption in recent years. However, existing studies often suffer from information leakage and rely on inappropriate hyperparameter tuning. To address these issues, we propose VMDNet, a causality-preserving framework that (i) applies sample-wise VMD to avoid temporal leakage; (ii) represents each decomposed mode with frequency-aware embeddings and decodes it using parallel temporal convolutional networks (TCNs), ensuring mode independence and efficient learning; and (iii) introduces a Stackelberg game inspired bilevel scheme to guide the selection of VMD’s two key hyperparameters. Experiments on three widely used electricity demand datasets show that VMDNet consistently outperforms state-of-the-art baselines.

Index Terms—Electricity demand forecasting, Variational Mode Decomposition (VMD), Bilevel optimization

I. INTRODUCTION

Accurate electricity demand forecasting is a fundamental task for the reliable and economical operation of modern power systems [1]. It directly supports generation scheduling, grid dispatch, and market clearing, and becomes increasingly challenging with the growing penetration of renewable energy sources [2]. Given the inherent dynamics and temporal dependencies of electric load data [3], accurate modeling of periodic patterns and multi-scale structures is critical for robust forecasting [4]. This motivates the incorporation of decomposition techniques, which aim to separate raw sequences into interpretable components that facilitate predictions.

VMD [5] is an advanced signal processing technique that decomposes a time series into a finite number of band-limited modes, also called intrinsic mode functions (IMFs), each centered on a distinct frequency. Unlike traditional tools such as Fourier or wavelet transforms, VMD enforces spectral sparsity and mode separation through variational optimization, producing non-overlapping modes with compact bandwidths. This leads to a more interpretable and disentangled representation of the original signal. Empirical evidence has demonstrated that it is effective in energy forecasting [6], [7].

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However, two issues remain and potentially prevent its practical application to forecasting: (i) the prevalent practice of decomposing the entire series before training grants implicit access to future information and violates causality [8]; and (ii) hyperparameter sensitivity, such that performance depends critically on the number of modes K and the bandwidth penalty α [9].

Regarding issue (i), some studies decompose the data after the train/test split to avoid cross-set leakage [10], [11]. Even so, the decomposition is still computed on entire input subsequences, which leaks future information. Moreover, short-window decomposition often yields weakly separated modes whose identities drift across samples, and the forecasting architectures used in these works are not designed to robustly represent such unstable modes.

For issue (ii), existing works mostly rely on meta-heuristic searches (e.g., GA, PSO) to optimise simple proxy measures or statistics, such as reconstruction error or sample entropy to raise decomposition quality [12], [13]. Nevertheless, gains under these proxies do not necessarily translate into better forecasting, because such proxies ignore the representational requirements of the downstream predictor. In addition, most previous studies typically treat K and α independently, despite their asymmetric importance.

To address these limitations in a unified and task-aware manner, we propose VMDNet, a leakage-free, multi-branch forecasting framework built upon variational mode decomposition. Our main contributions are summarised as follows.

- We propose a sample-wise VMD scheme to eliminate temporal information leakage in forecasting settings.
- Our framework employs a multi-branch decoding architecture with frequency-aware embeddings to model decomposed modes independently and robustly.
- A bilevel search motivated by the Stackelberg game is explored to provide task-aware guidance for selecting key VMD hyperparameters.

The implementation is released with a GPU-parallelised batch VMD module.¹

II. METHODOLOGY

A. Variational Mode Decomposition

VMD [5] decomposes a signal $f(t)$ into K band-limited IMFs $u_k(t)$, each centred at a frequency ω_k , by solving a constrained

¹<https://github.com/weibin-feng/VMDNet>

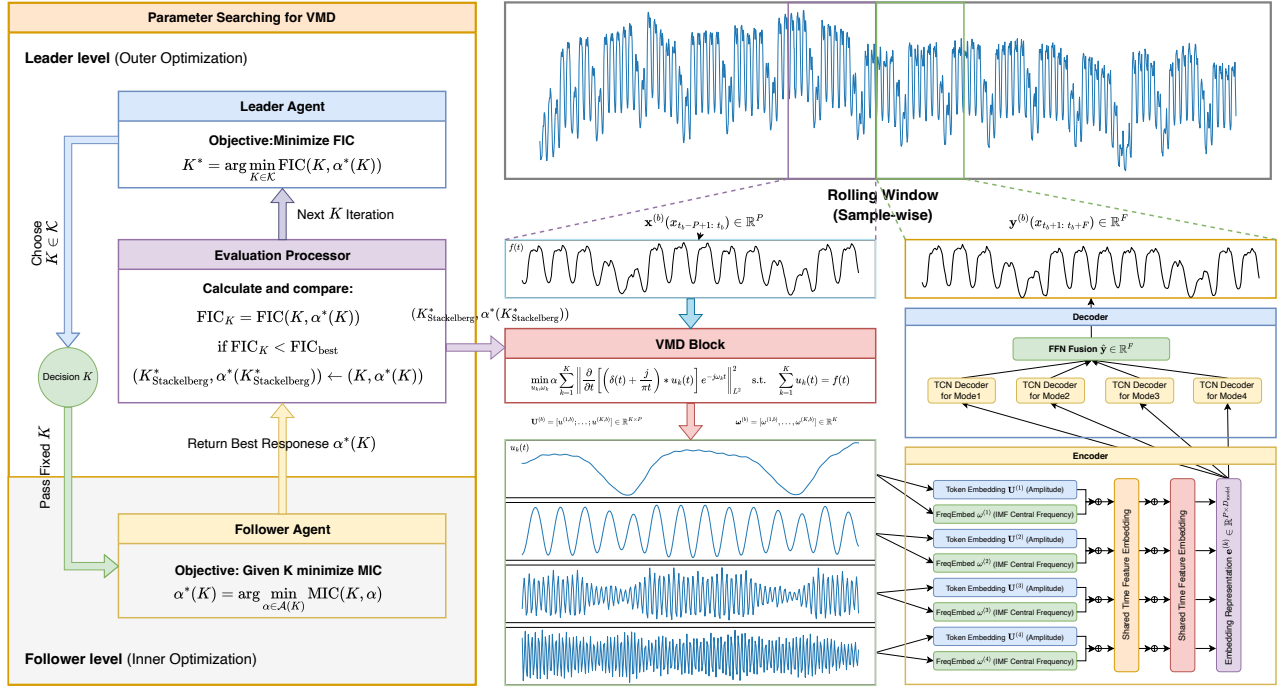


Fig. 1: Schematic of VMDNet. An example configuration on the ENTSO-E dataset with $K = 4$ is shown.

variational optimisation problem:

$$\min_{\{u_k\}, \{\omega_k\}} \left\{ \alpha \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_{L^2}^2 \right\} \quad (1)$$

subject to the reconstruction constraint

$$\sum_{k=1}^K u_k(t) = f(t). \quad (2)$$

Here, $\alpha > 0$ penalises the mode bandwidth, $*$ denotes the convolution, and $j^2 = -1$. The Hilbert transform $(\delta(t) + \frac{j}{\pi t}) * u_k(t)$ produces the analytic signal of $u_k(t)$; multiplying by $e^{-j\omega_k t}$ shifts it to baseband; ∂_t then measures the variation of this demodulated signal, and the L^2 norm quantifies the bandwidth around ω_k . The optimisation in Eqs. (1)–(2) is solved via ADMM.

B. Sample-wise Decomposition

Let the time series be $\mathcal{D} = \{x_t\}_{t=1}^T$. Given lookback length P and forecast horizon F , the forecasting problem is to learn a mapping

$$\mathcal{F}_\theta : \mathbb{R}^P \rightarrow \mathbb{R}^F, \quad \hat{\mathbf{y}}^{(b)} = \mathcal{F}_\theta(\mathbf{x}^{(b)}), \quad (3)$$

where $\mathbf{x}^{(b)} \in \mathbb{R}^P$ is the b -th input window and $\hat{\mathbf{y}}^{(b)} \in \mathbb{R}^F$ is the corresponding prediction.

For a stride $s \geq 1$, define the endpoint of the b -th window as $t_b = P + (b-1)s$. The input–target pair is then

$$\begin{aligned} \mathbf{x}^{(b)} &:= (x_{t_b-P+1:t_b}) \in \mathbb{R}^P, \\ \mathbf{y}^{(b)} &:= (x_{t_b+1:t_b+F}) \in \mathbb{R}^F. \end{aligned} \quad (4)$$

Stacking across $b = 1, \dots, B$ windows yields matrices $\mathbf{X} \in \mathbb{R}^{B \times P}$ and $\mathbf{Y} \in \mathbb{R}^{B \times F}$.

Each input window $\mathbf{x}^{(b)}$ is decomposed by a K -mode VMD operator with bandwidth penalty α :

$$\mathbf{U}^{(b)} = [u^{(1,b)}; \dots; u^{(K,b)}] \in \mathbb{R}^{K \times P}, \quad (5)$$

$$\boldsymbol{\omega}^{(b)} = [\omega^{(1,b)}, \dots, \omega^{(K,b)}] \in \mathbb{R}^K, \quad (6)$$

where $u^{(k,b)}$ denotes the k -th mode and $\omega^{(k,b)}$ its estimated center frequency. Each window satisfies the reconstruction constraint $\mathbf{x}^{(b)} = \sum_{k=1}^K u^{(k,b)}$.

Stacking across b gives

$$\mathbf{U} \in \mathbb{R}^{B \times K \times P}, \quad \boldsymbol{\Omega} \in \mathbb{R}^{B \times K}. \quad (7)$$

The decomposition on $\mathbf{x}^{(b)}$ depends only on $(x_{t_b-P+1:t_b})$, i.e., observations available at time t_b . Thus, samplewise decomposition preserves causality and avoids information leakage by design.

C. Encoding Architecture

Instead of applying heavy sequence encoders, we exploit the clean spectral structure of IMFs and adopt a simple embedding strategy as the encoder. For each sample $b = 1, \dots, B$, mode $k = 1, \dots, K$, and timestep $t = 1, \dots, P$, the embedded representation $\mathbf{e}_t^{(b,k)} \in \mathbb{R}^{d_{\text{model}}}$ is

$$\begin{aligned} \mathbf{e}_t^{(b,k)} &= \underbrace{\text{TokenEmbed}_k(u_t^{(k,b)})}_{\text{amplitude}} + \underbrace{\text{TimeEmbed}(t) + \text{PosEmbed}(t)}_{\text{shared}} \\ &\quad + \underbrace{\text{FreqEmbed}_k(\omega^{(k,b)})}_{\text{frequency}}. \end{aligned} \quad (8)$$

The embedding components are defined as follows:

- $\text{TokenEmbed}_k : \mathbb{R} \rightarrow \mathbb{R}^{d_{\text{model}}}$ is a mode-specific convolutional projection that captures local amplitude patterns of mode k .
- $\text{FreqEmbed}_k : \mathbb{R} \rightarrow \mathbb{R}^{d_{\text{model}}}$ projects the center frequency got from Eq. (7) into the embedding space via a learnable linear transformation.
- $\text{TimeEmbed}, \text{PosEmbed} : \mathbb{N} \rightarrow \mathbb{R}^{d_{\text{model}}}$ are shared across all modes and encode temporal features and relative position within the window.

Stacking across all samples, modes, and timesteps yields the representation tensor

$$\mathbf{E} \in \mathbb{R}^{B \times K \times P \times d_{\text{model}}}, \quad (9)$$

This embedding design introduces an explicit frequency domain inductive bias into the model, allowing each mode to carry knowledge of its spectral identity throughout the learning pipeline. Unlike standard token embeddings which focus solely on amplitudes, the proposed structure respects both temporal and spectral semantics of the decomposed signals.

To capitalise on the decomposition, we treat each mode as a distinct frequency scale and model it separately. Concretely, we use mode specific parameterisations for TokenEmbed_k and FreqEmbed_k , while sharing TimeEmbed and PosEmbed to preserve global temporal alignment. This specialisation preserves sub-frequency characteristics and mitigates cross mode interference (mode mixing), whereas globally shared weights would force heterogeneous sub-bands into a single representation and dilute mode specific information.

D. TCN-Based Parallel Decoding

We used a parallel decoding mechanism that assigns one TCN branch per mode. For mode k , the slice $\mathbf{E}^{(k)} \in \mathbb{R}^{B \times P \times d_{\text{model}}}$ is decoded as

$$\hat{\mathbf{y}}^{(k)} = \text{TCN}_k(\mathbf{E}^{(k)}) \in \mathbb{R}^{B \times F \times 1}. \quad (10)$$

Each $\text{TCN}_k(\cdot)$ consists of L residual blocks with dilated convolutions ($d = 2^\ell$, $\ell = 0 \dots L - 1$), causal padding, GELU activations, and dropout. This preserves causality and captures long-range dependencies. Outputs from all branches are stacked as

$$\hat{\mathbf{Y}} = \text{stack}(\hat{\mathbf{y}}^{(1)}, \dots, \hat{\mathbf{y}}^{(K)}) \in \mathbb{R}^{B \times K \times F \times 1}. \quad (11)$$

Finally, a lightweight feedforward network (FFN) aggregates mode-wise forecasts into the final prediction $\hat{\mathbf{y}} \in \mathbb{R}^{B \times F \times 1}$.

Symmetrically, the parallel decoding design follows the same principle of modeling each frequency scale separately. For strongly periodic, multi-scale series, sub-band signals are typically more predictable and locally stationary, so each branch can employ a lightweight causal decoder (dilated TCN) to achieve high accuracy with small capacity. Running K such decoders in parallel preserves the spectral separability of VMD modes, mitigates cross-mode interference, and yields efficient causal modeling. Finally, we fuse the branch-wise outputs with a two-layer MLP to produce the final forecast, thereby completing the decoding stage.

E. Parameter Searching for VMD

VMD requires tuning two key parameters: the number of modes K and the bandwidth penalty α . These parameters play asymmetric roles: K sets the decomposition resolution (dominant), whereas α refines spectral compactness within that resolution (secondary). To address this asymmetry, we adopt a Stackelberg game process. A Stackelberg game is a hierarchical framework with two agents in unequal roles: a leader moves first, anticipating the follower's best response; the follower observes the leader's decision and solves a conditional problem to respond optimally. The resulting outcome is a Stackelberg equilibrium, where neither side benefits from deviating given this sequence of moves [14], [15].

Instantiated for VMD, we define two objectives: (i) the Forecastability Information Criterion (FIC), which quantifies forecastability by fitting an $AR(r)$ model to each mode and summing the residual variances; and (ii) the Mutual Information Criterion (MIC), which quantifies cross-mode overlap via average pairwise mutual information. FIC additionally includes an AIC/BIC-inspired complexity penalty to discourage over-granular decompositions. Formally,

$$\begin{aligned} \text{FIC}(K, \alpha) = (T-r) \log \left(\sum_{k=1}^K \hat{\sigma}_k^2(K, \alpha) \right) \\ + (K(r+1)+1) \log(T-r). \end{aligned} \quad (12)$$

$$\text{MIC}(K, \alpha) = \frac{2}{K(K-1)} \sum_{i=1}^{K-1} \sum_{j=i+1}^K \hat{I}(u_i; u_j; K, \alpha). \quad (13)$$

Let $\mathcal{K} = \{K_{\min}, \dots, K_{\max}\}$ and $\mathcal{A}(K) = [\alpha_{\min}, \alpha_{\max}]$, this yields a Stackelberg bilevel search in Eqs. (14)–(15),

$$\alpha^*(K) = \arg \min_{\alpha \in \mathcal{A}(K)} \text{MIC}(K, \alpha), \quad (14)$$

$$K^* = \arg \min_{K \in \mathcal{K}} \text{FIC}(K, \alpha^*(K)). \quad (15)$$

Since the equilibrium is obtained via a heuristic without formal convergence guarantees, we perform multiple independent restarts with different random seeds, evaluate the resulting (K, α) pairs on the validation set, and select the best-performing pair as (K^*, α^*) .

III. EXPERIMENTS

A. Dataset

We evaluate the proposed model on three widely used electricity demand datasets [1] from (i) Poland power system demand (ENTSO-E)², (ii) ISO New England (ISO-NE)³, and (iii) Australian Energy Market Operator for New South Wales (AEMO)⁴. All datasets contain hourly records of average electricity demand and span the period from January 1, 2021 to June 13, 2024 (30,216 samples).

²<https://www.entsoe.eu/data/power-stats/>

³<https://www.iso-ne.com/isoexpress/web/reports/load-and-demand/>

⁴<https://www.aemo.com.au/energy-systems/electricity/>

TABLE I: Forecasting results at horizons $\{96, 192, 336\}$ with a lookback length of 336. Results are averaged over 5 runs with seeds $\{2021, 2022, 2023, 2024, 2025\}$. Each cell reports test MSE and MAE (mean $\pm$ std.); lower is better. Best and second best results are highlighted in **red bold** and blue underline, respectively.

Models	Metric	ENTSO-E			ISO-NE			AEMO		
		96	192	336	96	192	336	96	192	336
VMDNet	MSE	0.156 $\pm$ 0.011	0.217 $\pm$ 0.015	0.231 $\pm$ 0.007	<u>0.196 $\pm$ 0.009</u>	0.222 $\pm$ 0.007	0.216 $\pm$ 0.016	0.369 $\pm$ 0.005	0.414 $\pm$ 0.020	0.408 $\pm$ 0.005
	MAE	0.261 $\pm$ 0.007	0.304 $\pm$ 0.012	0.322 $\pm$ 0.011	<u>0.344 $\pm$ 0.010</u>	0.369 $\pm$ 0.006	0.363 $\pm$ 0.013	<u>0.426 $\pm$ 0.005</u>	0.455 $\pm$ 0.012	0.453 $\pm$ 0.005
TimeMixer [16]	MSE	0.173 $\pm$ 0.012	0.230 $\pm$ 0.008	0.287 $\pm$ 0.016	0.200 $\pm$ 0.005	0.238 $\pm$ 0.022	0.225 $\pm$ 0.011	0.411 $\pm$ 0.030	0.439 $\pm$ 0.014	<u>0.444 $\pm$ 0.023</u>
	MAE	0.275 $\pm$ 0.009	0.328 $\pm$ 0.008	0.372 $\pm$ 0.009	0.348 $\pm$ 0.005	0.380 $\pm$ 0.018	<u>0.368 $\pm$ 0.010</u>	0.439 $\pm$ 0.018	0.467 $\pm$ 0.008	0.473 $\pm$ 0.017
iTransformer [17]	MSE	0.182 $\pm$ 0.005	0.229 $\pm$ 0.010	0.402 $\pm$ 0.050	0.201 $\pm$ 0.005	0.233 $\pm$ 0.003	0.240 $\pm$ 0.005	0.381 $\pm$ 0.010	0.460 $\pm$ 0.016	0.542 $\pm$ 0.008
	MAE	0.290 $\pm$ 0.003	0.330 $\pm$ 0.008	0.447 $\pm$ 0.035	0.347 $\pm$ 0.005	0.379 $\pm$ 0.003	0.386 $\pm$ 0.004	0.433 $\pm$ 0.007	0.489 $\pm$ 0.011	0.542 $\pm$ 0.006
Crossformer [18]	MSE	0.277 $\pm$ 0.074	0.430 $\pm$ 0.131	0.555 $\pm$ 0.043	0.206 $\pm$ 0.007	0.253 $\pm$ 0.049	0.258 $\pm$ 0.043	0.382 $\pm$ 0.015	0.442 $\pm$ 0.012	0.482 $\pm$ 0.006
	MAE	0.363 $\pm$ 0.078	0.488 $\pm$ 0.119	0.580 $\pm$ 0.024	0.353 $\pm$ 0.007	0.395 $\pm$ 0.041	0.400 $\pm$ 0.034	0.431 $\pm$ 0.012	0.473 $\pm$ 0.010	0.498 $\pm$ 0.004
DLinear [19]	MSE	0.220 $\pm$ 0.002	0.267 $\pm$ 0.001	0.334 $\pm$ 0.005	0.198 $\pm$ 0.001	0.239 $\pm$ 0.000	0.223 $\pm$ 0.000	0.375 $\pm$ 0.000	<u>0.422 $\pm$ 0.000</u>	0.463 $\pm$ 0.000
	MAE	0.300 $\pm$ 0.003	0.328 $\pm$ 0.001	0.378 $\pm$ 0.006	<u>0.344 $\pm$ 0.000</u>	0.378 $\pm$ 0.001	0.371 $\pm$ 0.002	0.417 $\pm$ 0.000	<u>0.460 $\pm$ 0.000</u>	<u>0.470 $\pm$ 0.000</u>
PatchTST [20]	MSE	0.167 $\pm$ 0.003	0.226 $\pm$ 0.006	0.286 $\pm$ 0.004	0.211 $\pm$ 0.021	0.246 $\pm$ 0.010	0.278 $\pm$ 0.005	0.378 $\pm$ 0.004	0.459 $\pm$ 0.006	0.503 $\pm$ 0.014
	MAE	<u>0.267 $\pm$ 0.002</u>	0.314 $\pm$ 0.006	0.361 $\pm$ 0.003	0.355 $\pm$ 0.018	0.385 $\pm$ 0.009	0.414 $\pm$ 0.004	0.428 $\pm$ 0.002	0.476 $\pm$ 0.003	0.502 $\pm$ 0.008
TimesNet [21]	MSE	<u>0.163 $\pm$ 0.012</u>	<u>0.222 $\pm$ 0.035</u>	<u>0.274 $\pm$ 0.023</u>	0.187 $\pm$ 0.004	<u>0.232 $\pm$ 0.008</u>	<u>0.220 $\pm$ 0.007</u>	<u>0.372 $\pm$ 0.003</u>	0.431 $\pm$ 0.027	0.509 $\pm$ 0.014
	MAE	0.261 $\pm$ 0.011	<u>0.309 $\pm$ 0.025</u>	<u>0.353 $\pm$ 0.012</u>	0.325 $\pm$ 0.003	<u>0.374 $\pm$ 0.008</u>	<u>0.368 $\pm$ 0.006</u>	0.433 $\pm$ 0.005	0.464 $\pm$ 0.017	0.507 $\pm$ 0.013

B. Baselines

We perform a comprehensive comparison with six recent SOTA time series forecasting models: TimeMixer [16], iTransformer [17], Crossformer [18], DLinear [19], PatchTST [20], TimesNet [21]. Forecast accuracy is evaluated using mean squared error (MSE) and mean absolute error (MAE).

C. Experiment Setup

All models are trained with a fixed lookback length $P = 336$ and evaluated at horizons $F \in \{96, 192, 336\}$. The initial learning rate of 10^{-3} with Adam optimizer is adopted, train 10 epochs with early stopping. The batch size is set to 64 and the embedding dimension is 64. For parameter selection, we search $\mathcal{K} = \{2, 3, \dots, 15\}$ and $\mathcal{A}(K) = [500, 10000]$, and use AR order $p = 2$ when computing FIC, and perform 20 independent runs. All hyperparameters are tuned on the validation set; minor adjustments across horizons are applied. Training and evaluation are conducted on $4 \times$ NVIDIA GH200 (120 GB) Grace Hopper Superchips (CUDA 12.7) [22].

D. Results and Discussions

1) *Results Analysis:* As summarised in Table I, across 9 experimental settings spanning different datasets and forecasting horizons, VMDNet achieves the best results in 8 settings and the second-best result in the remaining one, demonstrating consistent superiority over strong baselines. Moreover, VMDNet maintains robust performance across datasets from different countries, indicating strong generalisation under diverse demand patterns.

TimesNet [21] achieves the second-best overall performance across the evaluated settings, highlighting the importance of frequency-oriented inductive biases through multi-period unfolding and 2D tensorization that capture periodic structures at a macro level. Performance differences across datasets further reveal how dataset characteristics interact with such

inductive biases. ENTSO-E and ISO-NE aggregate demand at the country/region level, yielding more stable periodic patterns, which particularly benefit periodicity-aware models such as TimesNet. In contrast, AEMO only covers New South Wales, where local demand is less averaged and exhibits higher variability, making linear models like DLinear more competitive. Nevertheless, VMDNet remains consistently strong across all datasets. This is because VMD decomposes the signal into band-limited modes via a global optimization objective, with each mode characterised by a well-separated center frequency. Such a frequency domain representation tends to be less sensitive to local noise and short-term fluctuations, as high-frequency noise is largely isolated into high-frequency components or attenuated, yielding more stable frequency embeddings for robust pattern learning. Compared with TimesNet’s high-level periodic modeling, VMDNet explicitly aligns representations in the frequency domain, leading to superior overall performance across diverse demand patterns.

2) *Ablation Study:* We conduct ablation studies to quantify the contribution of each component: (i) removing VMD by forecasting directly from raw inputs with a single TCN decoder; (ii) removing frequency encoding; (iii) removing parallel decoding by mean-aggregating all mode embeddings before a unified TCN decoder; (iv) removing both frequency encoding and parallel decoding; and (v) replacing the proposed bilevel parameter selection scheme with a PSO baseline that minimizes envelope entropy. All variants are trained and evaluated on ENTSO-E with horizon $F=336$.

As shown in Table II, compared with the full model, removing VMD results in an MSE increase of 0.018 (7.8%), highlighting the importance of frequency-aware decomposition. Frequency encoding also contributes consistently, as its removal increases the MSE by 0.013 (5.6%). Among all components, parallel decoding plays the most critical role, causing

TABLE II: Ablation results on the ENTSO-E dataset.

Models / Variants	ENTSO-E (336)	
	MSE	MAE
VMDNet (full)	0.231 ± 0.007	0.322 ± 0.011
w/o VMD	0.249 ± 0.004	0.339 ± 0.006
w/o frequency encoding	0.244 ± 0.010	0.334 ± 0.002
w/o parallel decoding	0.274 ± 0.027	0.353 ± 0.023
w/o freq. enc. & par. dec.	0.287 ± 0.017	0.357 ± 0.007
w/o bilevel searching	<u>0.240 ± 0.009</u>	<u>0.329 ± 0.010</u>

the largest performance drop (0.043, 18.6%) when removed. When both frequency encoding and parallel decoding are discarded, the degradation is further amplified to 0.056 (24.2%), demonstrating their complementary effects. In contrast, bilevel searching provides a moderate gain (0.009, 3.9%), serving as a performance booster rather than a dominant factor.

3) *Computational Cost*: Overall, the additional cost mainly comes from the bilevel parameter search and the VMD preprocessing. The bilevel search incurs extra offline cost: 30 restarts take 556.2 s for our method versus 385.9 s for the PSO baseline, yielding a 3.9% performance gain, and is conducted only once before training. At inference time, VMD accounts for most of the additional runtime (1.7180 ms per sample on average). Nevertheless, thanks to the lightweight encoder and decoder, the overall inference time remains low. Our method ranks 4th in inference speed among all compared models (1.8519 ms per sample on average), providing a favourable accuracy–efficiency trade-off with consistent performance gains (up to 7.8%).

IV. CONCLUSION

We introduced VMDNet, an information leakage-free VMD based model for electricity demand forecasting that integrates sample-wise variational mode decomposition with frequency-aware representations and parallel temporal convolutional decoding. By explicitly modeling decomposed modes in the frequency domain and guiding hyperparameter selection via a bilevel Stackelberg game, VMDNet effectively exploits multi-periodic structures in real-world demand series while avoiding information leakage. Experiments on three real-world electricity demand datasets demonstrate that VMDNet consistently outperforms strong baselines across multiple forecasting horizons and evaluation metrics. Future extensions may explore multivariate forecasting settings and learning differentiable neural network–based variants of VMD to enable end-to-end gradient-based optimisation of decomposition parameters.

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REFERENCES

[1] Q. Dong, R. Huang, C. Cui, D. Towey, L. Zhou, J. Tian, and J. Wang, “Short-term electricity-load forecasting by deep learning: A comprehensive survey,” *Engineering Applications of Artificial Intelligence*, vol. 154, p. 110980, 2025. [Online]. Available: <https://doi.org/10.1016/j.engappai.2025.110980>

[2] X. Wu, Y. Wang, Y. Bai, Z. Zhu, and A. Xia, “Online short-term load forecasting methods using hybrids of single multiplicative neuron model, particle swarm optimization variants and nonlinear filters,” *Energy Reports*, vol. 7, pp. 683–692, 2021. [Online]. Available: <https://doi.org/10.1016/j.egy.2021.01.030>

[3] A. Haque and S. Rahman, “Short-term electrical load forecasting through heuristic configuration of regularized deep neural network,” *Applied Soft Computing*, vol. 122, p. 108877, 2022. [Online]. Available: <https://doi.org/10.1016/j.asoc.2022.108877>

[4] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, “Informer: Beyond efficient transformer for long sequence time-series forecasting,” in *35th AAAI*, vol. 12B. AAAI, 2021, pp. 11 106–11 115.

[5] K. Dragomiretskiy and D. Zosso, “Variational mode decomposition,” *IEEE Transactions on Signal Processing*, vol. 62, pp. 531–544, 2 2014. [Online]. Available: <https://doi.org/10.1109/TSP.2013.2288675>

[6] K. Ma, X. Nie, J. Yang, L. Zha, G. Li, and H. Li, “A power load forecasting method in port based on VMD-ICSS-hybrid neural network,” *Applied Energy*, vol. 377, p. 124246, 2025. [Online]. Available: <https://doi.org/10.1016/j.apenergy.2024.124246>

[7] H. Han, J. Peng, J. Ma, S. L. Liu, and H. Liu, “Research on load forecasting based on CEEMDAN SE VMD and SelfAttention TCN fusion model,” *Scientific Reports*, vol. 15, no. 1, p. 14530, Apr. 2025. [Online]. Available: <https://doi.org/10.1038/s41598-025-98224-4>

[8] Z. Qian, Y. Pei, H. Zareipour, and N. Chen, “A review and discussion of decomposition-based hybrid models for wind energy forecasting applications,” *Applied Energy*, vol. 235, pp. 939–953, 2019. [Online]. Available: <https://doi.org/10.1016/j.apenergy.2018.10.080>

[9] V. Krishna Rayi, S. P. Mishra, J. Naik, and P. K. Dash, “Adaptive VMD based optimized deep learning mixed kernel ELM autoencoder for single and multistep wind power forecasting,” *Energy*, vol. 244, 4 2022.

[10] L. Chen and X. Zhao, “A multiscale and multivariable differentiated learning for carbon price forecasting,” *Energy Economics*, vol. 131, p. 107353, 2024. [Online]. Available: <https://doi.org/10.1016/j.eneco.2024.107353>

[11] Y. Huang, X. Dai, Q. Wang, and D. Zhou, “A hybrid model for carbon price forecasting using GARCH and long short-term memory network,” *Applied Energy*, vol. 285, p. 116485, 2021. [Online]. Available: <https://doi.org/10.1016/j.apenergy.2021.116485>

[12] K. Choudhary, G. K. Jha, R. Jaiswal, and R. R. Kumar, “A genetic algorithm optimized hybrid model for agricultural price forecasting based on VMD and LSTM network,” *Scientific Reports*, vol. 15, no. 1, p. 9932, Mar. 2025. [Online]. Available: <https://doi.org/10.1038/s41598-025-94173-0>

[13] G. An, L. Chen, J. Tan, Z. Jiang, Z. Li, and H. Sun, “Ultra-short-term wind power prediction based on PVMD-ESMA-DELM,” *Energy Reports*, vol. 8, pp. 8574–8588, 2022. [Online]. Available: <https://doi.org/10.1016/j.egy.2022.06.079>

[14] H. Von Stackelberg, *Market structure and equilibrium*. Springer Berlin Heidelberg, 2011.

[15] B. Colson, P. Marcotte, and G. Savard, “An overview of bilevel optimization,” *Annals of Operations Research*, vol. 153, pp. 235–256, 9 2007. [Online]. Available: <https://doi.org/10.1007/s10479-007-0176-2>

[16] S. Wang, H. Wu, X. Shi, T. Hu, H. Luo, L. Ma, J. Y. Zhang, and J. Zhou, “TimeMixer: Decomposable multiscale mixing for time series forecasting,” in *12th ICLR*. ICLR, 2024.

[17] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, and M. Long, “iTransformer: Inverted transformers are effective for time series forecasting,” in *12th ICLR*. ICLR, 2024.

[18] Y. Zhang and J. Yan, “Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting,” in *ICLR*, 2023.

[19] A. Zeng, M. Chen, L. Zhang, and Q. Xu, “Are transformers effective for time series forecasting?” in *37th AAAI*, vol. 37. AAAI Press, 6 2023, pp. 11 121–11 128.

[20] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *11th ICLR*. ICLR, 2023.

[21] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, “TimesNet: Temporal 2D-variation modeling for general time series analysis,” in *11th ICLR*. ICLR, 2023.

[22] S. McIntosh-Smith, S. Alam, and C. Woods, “Isambard-AI: a leadership-class supercomputer optimised specifically for Artificial Intelligence,” in *Cray User Group, CUG 24*, 2025, p. 44–54. [Online]. Available: <https://doi.org/10.1145/3725789.3725794>